

Q Reverse Array (GFG)

Brute force :- How we can make a new arr & store the value from the end.

Ex:- [5, 8, 12, 16]

new Arr [16, 12, 8, 5]
[5, 8, 12, 16]

Pseudo Code:-

- ① int [] arr = new int[n];
- ② int i = 0;
- ③ for (int j = n-1; j >= 0; j--) {
- ④ arr[i] = arr[j];
- ⑤ i++;
- ⑥ }
- ⑦ return arr;

Problem:-

It uses extra space O(n)

Optimal way:- Two pointer + Swap. T.C $\rightarrow$ O(n) S.C $\rightarrow$ O(1)

Ex:- [5, 6, 8, 10, 12]

[5, 6, 8, 10, 12]
[12, 10, 8, 6, 5]
[12, 10, 8, 6, 5] $\rightarrow$ Here we break the loop.

Pseudo Code

- ① start = 0
- ② end = n-1;
- ③ while (start < end) {
- ④ swap(arr[start], arr[end]);
- ⑤ start++;
- ⑥ end--;
- ⑦ }